

RESEARCH GRANT PROPOSAL

SAFE-RAG: A Schema-Grounded and Faithfulness-Aware Self-Correcting Retrieval-Augmented Generation Framework for Trustworthy Financial Compliance Question Answering

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Proposed Duration	12 months
Submission Date	July 2026

1. Executive Summary

Retrieval-augmented generation (RAG) has become the default approach for question answering over financial filings, regulatory rulebooks, and audit documentation, because it grounds model output in retrieved evidence rather than relying on parametric memory alone. In financial compliance settings, however, a free-text answer is rarely sufficient: a compliance officer or downstream reporting system needs a structured judgment, for example a clause identifier, an obligation status, and a supporting citation, that can be parsed without further human interpretation.

A scoping review of 35 papers (2021-2026) conducted for this proposal, spanning schema-constrained generation, RAG faithfulness evaluation, agentic self-correction, and financial document question answering, found that no existing system jointly performs schema validation, evidence-faithfulness checking, and typed corrective routing in a financial or regulatory compliance setting. Thirty of those papers were identified in the original May-July 2026 search; a further five were added by a July 2026 update sweep of arXiv listings, and three of the five materially narrow the gap claimed by an earlier draft of this proposal, as recorded in Sections 3, 4 and 8.1. Systems that guarantee schema-valid output make no claim about whether the values inside that output are actually supported by evidence; systems that check faithfulness are evaluated almost exclusively on free text and cannot detect deceptive grounding, a documented failure mode in which a response is faithful to some retrieved passage but wrong because that passage concerns the wrong entity or clause.

This project proposes SAFE-RAG, a framework that classifies each generated answer as schema-invalid, faithful-but-misattributed, or valid, and routes each condition to a distinct corrective action: local regeneration, entity-constrained re-retrieval, or abstention. The typed-routing paradigm the design adopts was first demonstrated in a tool-calling agent setting, where it raised recovery rates from 32.76% to 89.68% [11], and was carried into retrieval-augmented generation during 2026 [19], [20], [21]. That typed routing outperforms undifferentiated retry is therefore taken here as an established premise rather than claimed as a finding of this project. What remains untested, and what this proposal claims, is the pair of conditions being routed over: every published RAG diagnosis vocabulary is trajectory-oriented and can express neither schema invalidity against a target output schema nor entity-level misattribution, and none of this work has been evaluated on financial or regulatory content. SAFE-RAG will be evaluated

against pre-registered, falsifiable hypotheses on the ObliQA regulatory corpus and FinanceBench-derived financial filings, under a correction budget matched across every compared condition. The project is scoped to deliver a publishable contribution within 12 months, with an initial deployment context aligned to Malaysia’s regulatory-technology development priorities. [18]

2. Background

Financial institutions increasingly rely on large language models to support regulatory compliance and document analysis. Retrieval-augmented generation improves factual grounding over parametric generation alone and has become the default architecture for question answering over structured financial and regulatory text, building on an established benchmark landscape of numerical and tabular reasoning datasets such as FinQA [1], TAT-QA [2], ConvFinQA [3], and MultiHiertt [4], and alongside finance-specialized language models such as BloombergGPT [5].

Two distinct failure surfaces nonetheless remain unresolved in this setting. First, generated output can fail to conform to the structured schema a downstream compliance system expects. Second, output can conform to that schema while still being unsupported by, or misattributed to, the wrong part of the retrieved evidence. Existing research addresses these two surfaces in separate literatures: work on schema-constrained decoding such as XGrammar [6] guarantees syntactic validity but makes no claim about evidential correctness, while work on faithfulness and hallucination detection such as RAGAS [7], AlignScore [8], and FRANQ [9] checks whether a claim is entailed by retrieved context but does not verify that the evidence concerns the correct entity, clause, or category.

3. Problem Statement

Current RAG systems deployed in financial and regulatory compliance settings cannot reliably distinguish between two failure modes: **structural failure**, where the output does not parse against the required schema, and **attribution failure**, where the output is schema-valid and passes standard faithfulness checks, yet cites evidence about the wrong entity, clause, or category. This second failure mode, recently termed *deceptive grounding* [10], is invisible to standard faithfulness metrics because those metrics ask only whether a claim is supported by some evidence, not whether that evidence is about the correct subject.

Compounding this, systems that do detect a failure typically respond with the same generic retry action regardless of what actually went wrong. That typing a failure before correcting it outperforms generic retry is, however, no longer an open question. It was first demonstrated on a tool-calling agent benchmark, where typed, failure-specific correction achieved 89.68% recovery against 32.76% for generic retry [11], and during 2026 the same paradigm was carried into retrieval-augmented generation. One system diagnoses the failure type and the earliest failure point in an agentic RAG trajectory and then applies a repair conditioned on that diagnosis, reporting higher answer accuracy at lower token cost than rerun-style recovery [19]; a second reaches a comparable result by routing on a probed hidden-state failure state [20]; and a third evaluates diagnose-and-repair under an explicitly fixed correction budget [21]. This proposal therefore adopts typed routing as a premise of its design rather than presenting it as a finding.

What remains unresolved is the vocabulary in which failures are diagnosed. Every existing RAG diagnosis vocabulary is trajectory-oriented, consisting of categories such as retrieval insufficiency, reasoning error, and malformed tool call. None diagnoses schema validity against a target output schema, and none per-

forms entity-level attribution checking. The second omission is not incidental. A deceptive-grounding failure is, by construction, a trajectory in which retrieval succeeded, reasoning was sound, and every claim is entailed by retrieved text, so it is invisible to a sufficiency-based diagnosis model for exactly the reason it is invisible to a faithfulness metric [10]. A diagnosis vocabulary built for trajectory repair cannot express it. The closest available vocabulary is a peer-reviewed taxonomy of the error types that occur in realistic RAG systems, released with an annotated error dataset and a taxonomy-aligned automatic evaluator [22], but that work stops at classification and offline evaluation and does not drive an inference-time controller. No system identified in this review applies typed failure classification to the specific combination of schema-validity and evidence-faithfulness failures that arises in structured-output RAG for financial and regulatory compliance.

4. Research Gap

A capability coverage analysis of the 35 reviewed studies, scored against five dimensions, schema-constrained structured output (C1), faithfulness and evidence-grounding (C2), agentic retrieval control (C3), typed error routing (C4), and financial or regulatory compliance domain application (C5), returns two findings. The first is a correction. C4 is no longer empty outside the tool-calling domain: three of the five studies added by the July 2026 update sweep score full coverage on typed error routing within retrieval-augmented generation [19], [20], [21]. An earlier draft of this proposal claimed that C4 was unoccupied in the RAG setting, and that claim is withdrawn. The second finding is unaffected by those additions: no reviewed study scores full coverage on C1, C2 and C4 simultaneously. The one reviewed system that couples schema structure with evidence attribution does so in the regulatory domain but has no typed routing mechanism [23]; the three RAG typed-routing systems implement neither schema validation nor entity-level attribution checking; and the original typed-routing system attains C4 outside RAG entirely [11]. That empty three-way conjunction, rather than typed routing in general, is what this proposal claims.

Three concrete gaps follow from this pattern, and they are complementary rather than independent:

- **Gap 1:** schema validity and faithfulness are addressed by largely disjoint literatures that rarely cite each other and have not been combined into a single validation step that checks both properties of the same output. The single exception among the reviewed studies enforces a per-rule schema by injecting it at generation time in order to make attribution checkable in regulatory compliance question answering [23]; because that constraint is applied uniformly to every query, the two properties are never diagnosed separately as conditions of the same generated output.
- **Gap 2:** typed error routing outperforms generic retry, and this has now been shown inside retrieval-augmented generation as well as for tool-calling agents [11], [19], [20], [21]. What has not been shown is routing over the pair of conditions targeted here. Every published diagnosis vocabulary types failures at the level of the agent trajectory; none diagnoses schema invalidity against a target output schema, none checks whether cited evidence concerns the entity, clause, or category named in the query, and none of this work is evaluated on financial or regulatory content.
- **Gap 3:** financial RAG systems invest heavily in retrieval engineering, fine-tuned retrievers, hybrid search, reranking, but remain single-pass at inference, with no per-query, quality-triggered repair loop. The closest competitor intervenes on attribution in the regulatory domain by constraining generation uniformly rather than by repairing diagnosed failures [23], so it never has to decide which condition failed: it has no schema-invalidity branch, no re-retrieval branch, no abstention

branch, and it reports no comparison against an undifferentiated retry baseline.

Together, these gaps motivate a framework that diagnoses schema validity and evidence faithfulness as two separately detected conditions, transplants typed routing onto that specific pair of conditions, and targets financial compliance question answering specifically, where institutional demand for structured, auditable output is already documented [15], [16] and existing benchmarks remove the need to construct a new corpus from scratch.

5. Research Questions

RQ1. Can a schema validator and an evidence-attribution checker operate as two independently diagnosable conditions over structured RAG output in the financial compliance domain, such that each condition can be detected without relying on the other?

RQ2. Given that typed routing is already established to outperform undifferentiated retry over trajectory-level failure taxonomies [11], [19], [20], does the specific two-condition typing proposed here, schema invalidity routed to local regeneration and evidence misattribution routed to entity-constrained re-retrieval, outperform an undifferentiated single-retry baseline that uses the same validator and checker components, under a matched correction budget?

RQ3. Does the proposed evidence-attribution check reduce deceptive-grounding-style errors, defined as schema-valid, faithfulness-metric-passing answers that cite evidence concerning the wrong entity, clause, or category, relative to standard faithfulness checking alone?

6. Research Objectives

RO1. Complete a PRISMA 2020-compliant systematic literature review that extends the 35-paper scoping review already conducted for this proposal, and finalize the schema design and adversarial test subset.

RO2. Develop a schema validator and an evidence-attribution checker that independently diagnose structural and evidential failure in generated RAG output.

RO3. Develop a typed corrective-routing controller that maps each diagnosed failure condition to a distinct action: local regeneration, entity-constrained re-retrieval, or abstention.

RO4. Implement and evaluate SAFE-RAG against five baseline conditions and six metrics on the ObliQA and FinDER/FinanceBench corpora, with every condition run under a matched correction budget, and test pre-registered hypotheses H1-H3.

7. Scope of Research

The project is scoped to financial and regulatory compliance document question answering. The primary evaluation corpus is ObliQA [15], comprising 27,869 question-and-passage pairs drawn from real Abu Dhabi Global Markets financial regulation, supplemented by FinDER [16] (5,703 expert-annotated analyst-style query, evidence, and answer triplets drawn from S&P 500 filings) and by filings from the FinanceBench benchmark [17]. A subset of both corpora will be extended into schema-bound question-and-answer pairs, and a smaller adversarial test set will be hand-constructed to contain deceptive-grounding-style failures, which neither corpus contains by design.

The scope explicitly excludes construction of a new base corpus, and excludes empirical demonstration of domain generality beyond finance; the latter is identified as a direction for future work rather than a claim of the present project.

8. Research Methodology

8.1 Systematic Review

The proposal is grounded in a scoping review already conducted (May-July 2026) using targeted Boolean searches across arXiv, Google Scholar, ACL Anthology, and IEEE Xplore, from which 30 papers meeting defined inclusion criteria were selected for full-text review. A supplementary update sweep was run in July 2026 against arXiv listings for cs.CL and cs.IR, using the additional query terms (“failure-aware” OR “diagnose” OR “repair”) AND (“retrieval-augmented generation” OR “agentic RAG”), specifically to capture typed-routing work published into the RAG setting during the review period. That sweep returned five further papers meeting the same inclusion criteria [19], [20], [21], [22], [23], bringing the total to 35. Three of the five materially narrow the gap claimed by an earlier draft of this proposal, and their inclusion is reported explicitly here rather than absorbed silently, because the publication rate in this area is itself a threat to the validity of any gap claim (Section 8.4).

The funded project will formalize this scoping work into a complete, reproducible systematic review following the PRISMA 2020 reporting standard and the Kitchenham and Charters protocol for software-engineering literature reviews, including full per-database screening counts and a supervisor-verified second screening pass to address the single-reviewer limitation of the scoping stage.

8.2 SAFE-RAG Architecture

SAFE-RAG extends a standard retrieve-then-generate pipeline with two independently diagnosed checks and a routing function. A hybrid retriever combines dense retrieval scores in the style of Dense Passage Retrieval [13] with sparse lexical scores in the style of BM25 [14], optionally followed by cross-encoder reranking, to return the top-k passages for a query. A generator then produces a structured answer conditioned on the query, the retrieved context, and a target schema.

A schema validator computes a binary pass/fail indicator of whether the generated answer parses against the schema. On failure, the corrective action is local regeneration against the same retrieved context, since the fault concerns output structure rather than evidence quality. On schema pass, an evidence-attribution checker computes a composite score combining an entailment component, following established faithfulness-scoring approaches [7], [8], with an attribution component that verifies the retrieved evidence concerns the same entity, clause, or category named in the query, following the entity-attribution verification approach used to detect deceptive grounding [10]. A routing function uses both checks, together with the number of correction attempts already made, to decide whether to return the answer, regenerate, re-retrieve using an entity-constrained reformulated query, or abstain after a configured maximum number of attempts.

Two considerations justify diagnosing the conditions separately rather than collapsing them into a single quality score. First, self-correction driven by no external, specific signal of what is wrong frequently fails to improve output and can degrade it [12]. Second, reparability depends on failure type: format failures can be repaired by rewriting over evidence that is already sufficient, whereas evidence failures cannot, an asymmetry reported directly in the RAG repair literature [19]. A two-condition router is the mechanism

by which that asymmetry is exploited.

8.3 Evaluation Design

Five conditions will be compared: naive RAG with no schema enforcement and no faithfulness check (a floor); schema-constrained generation only; faithfulness-checking only; an undifferentiated single-retry baseline in which any detected failure triggers the same generic retry regardless of type, using the same underlying validator and checker components (the key comparison for RQ2); and the full SAFE-RAG system.

All five conditions will be run under a matched correction budget, defined as an equal maximum number of generator and retriever calls per query. This control is not optional: a repair loop compared against a single-pass baseline without budget control measures additional compute rather than the value of the diagnosis [21], and results will not be reported in a form that confounds the two. A secondary sweep will vary the maximum number of correction attempts across one, two, and three to characterize how the advantage of typing scales with the budget available.

Six metrics will be reported: schema validity rate; faithfulness score (entailment-based, consistent with RePASs and AlignScore [8]); entity-attribution accuracy, a metric introduced for this project; end-task correctness; repair efficiency (additional model or retrieval calls required per query, within the matched budget); and abstention precision/recall. To avoid the circularity of using a single automated judge both to trigger repair and to score the result, the entailment model used inside the routing loop and the model used to compute the reported faithfulness metric will be drawn from different model families, and both will be validated against a human-annotated subsample of at least 150 items before the main experiments are run.

Three pre-registered hypotheses will be tested:

- **H1.** On the adversarial deceptive-grounding test subset, SAFE-RAG achieves entity-attribution accuracy at least 15 percentage points higher than the faithfulness-checked-only baseline.
- **H2.** Under a matched correction budget, SAFE-RAG achieves higher end-task correctness than the undifferentiated single-retry baseline while consuming no more additional model or retrieval calls per query on average. The budget-matching condition is what makes this hypothesis falsifiable; without it, any observed difference is attributable to additional compute [21].
- **H3.** Ablating the typed-routing step, so any detected failure triggers the same generic retry action using the same validator and checker components, reduces end-task correctness relative to the full SAFE-RAG system, mirroring the ablation result reported for typed routing in its original tool-calling domain [11] and its replication within RAG [19].

8.4 Limitations and Risk Mitigation

Two risks specific to this proposal are recorded here, alongside the review limitations noted in Section 8.1, so that their mitigations form part of the funded work plan.

The first concerns the durability of the gap claim itself. A contribution defined by an absence in a fast-moving literature decays measurably over months: three of the five papers added by the July 2026 update sweep were posted within the four months preceding this submission, and two of them narrowed the gap claimed by an earlier draft of this document. The mitigation is procedural rather than analytical. The

claim is stated at the narrowest defensible scope rather than the broadest available one, restricted to the conjunction of schema invalidity and evidence misattribution in a compliance setting; a standing arXiv alert on the four query clusters used in the review will be maintained for the duration of the project; and the capability coverage assessment described in Section 4 will be re-run at each milestone review. A gap claim in this area should be read as a dated observation, not as a stable property of the literature.

The second concerns the base rate of the failure mode the attribution branch targets. Because the adversarial subset described in Section 7 is hand-constructed, its distribution does not by itself establish how often deceptive grounding occurs in practice. The rate at which it arises naturally in ObliQA and FinDER will therefore be measured and reported as a descriptive statistic in its own right, and a low measured rate would weaken the practical motivation for the attribution branch even if the H1 result holds on the adversarial subset. That measurement is scheduled early, in Months 1 to 2, precisely so that a negative result arrives before the implementation phase commits resources, allowing the balance of effort between the schema-validity and attribution branches to be revised while the project can still absorb the change.

9. Expected Outcomes

- A validated SAFE-RAG framework and an open research prototype comprising the schema validator, evidence-attribution checker, and typed routing controller.
- A completed PRISMA 2020-compliant systematic review formalizing the present 35-paper scoping review.
- A reusable adversarial deceptive-grounding test set for financial and regulatory RAG evaluation, together with a reported measurement of the natural rate of deceptive grounding in ObliQA and FinDER.
- A manuscript reporting the framework and evaluation results, targeted at a regulatory natural language processing or applied machine learning venue, with submission planned before the end of the project.
- A direct contribution toward the applicant's graduate research program, extending the applicant's prior schema-validated self-correction prototype into a formally evaluated contribution.

10. Research Contributions

This project contributes: (1) a 35-paper scoping literature review scored against five capability dimensions, establishing that the conjunction of schema-constrained output, evidence-attribution checking, and typed corrective routing is currently unoccupied, while recording explicitly that typed routing on its own is now established within RAG; (2) SAFE-RAG, an architecture that performs schema validation and evidence-attribution checking as two independently diagnosed conditions and routes each detected failure type to a distinct corrective action; (3) an evaluation plan with pre-registered, falsifiable hypotheses, run under a matched correction budget and built on existing regulatory and financial benchmarks rather than a newly constructed corpus; and (4) a concrete deployment pathway aligned with Malaysia's regulatory-technology development priorities [18].

11. Project Timeline

The timeline below is deliberately compressed to 12 months rather than the multi-year horizon typical of a new framework proposal. This is possible because the scoping review, gap analysis, and architecture design described in this proposal are already complete; the funded period begins directly with implementation. The literature review, framework design, and evaluation plan are treated as parallel, continuously updated work rather than sequential gates, so that a submittable manuscript can be assembled from Month 9 onward rather than only after every phase below has fully closed. The literature-monitoring task described in Section 8.4 runs across the full period rather than concluding with the Month 2 review.

Phase	Months	Milestone / Deliverable
1. Systematic review & design	1–2	Full PRISMA 2020-compliant systematic review completed (extending the 35-paper scoping review already conducted for this proposal); schema design finalized; adversarial deceptive-grounding test subset constructed; natural rate of deceptive grounding in ObliQA and FinDER measured and reported.
2. Core diagnostic components	3–5	Schema validator and evidence-attribution checker implemented and unit-validated against a human-annotated subset; entailment models for the routing loop and for the reported faithfulness metric selected from different model families.
3. Typed routing controller	6–8	Rule-based routing policy implemented and integrated with the diagnostic components; learned routing policy explored if time permits.
4. Evaluation	9–10	All five baseline conditions run on ObliQA and FinDER/FinanceBench under a matched correction budget; hypotheses H1–H3 tested against pre-registered criteria.
5. Analysis & publication	11–12	Results analysed; manuscript prepared and submitted to a regulatory NLP or applied machine learning venue.

12. Conclusion

This proposal identifies a specific, evidenced gap in the current literature. Typed error routing, established first for tool-calling agents and carried during 2026 into agentic RAG trajectories, is a demonstrated mechanism rather than an open question, and this proposal treats it as a premise. What no reviewed system does is diagnose output structure and evidence attribution as two independently detectable conditions over the same structured output and route each to a corrective action selected on the basis of which condition failed. The existing diagnosis vocabularies, built for trajectory-level repair, cannot express a faithful-but-misattributed answer at all, and no typed-routing system in this line has been evaluated on financial or regulatory content. SAFE-RAG addresses that gap directly, diagnosing schema invalidity and evidence misattribution as two independently detected conditions and routing each to a distinct corrective action, while building on existing regulatory and financial benchmarks rather than requiring new data collection.

The evaluation plan specifies falsifiable, pre-registered hypotheses against both an undifferentiated-retry baseline and a faithfulness-only baseline, under a correction budget matched across all conditions, directly testing whether typed diagnosis over this particular pair of conditions, rather than correction activity or additional compute in general, accounts for any observed improvement. Within a compressed 12-month timeline, the project is designed to extend the applicant's prior schema-validated self-correction prototype into a formally evaluated, publishable research contribution, with an initial deployment context aligned to Malaysia's regulatory-technology priorities and a technical contribution intended to gen-

eralize to other high-stakes, structured-output domains in future work.

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